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Boosted Trumpet Initial Data

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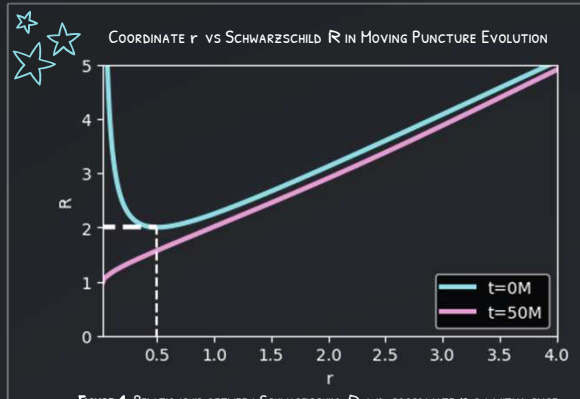


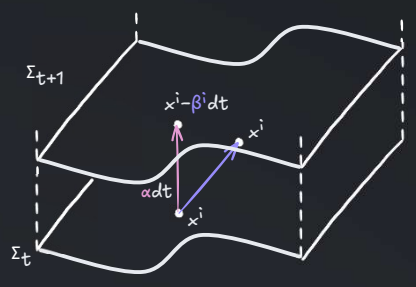
Figure 1: Relationship between Schwarzschild R and coordinate r on initial slice t=0 and then at t=50M. The t=0 hypersurface represents a wormhole (there are two copies of the R>2M region). The t=50M slice represents a trumpet, where r=0 corresponds to a finite Schwarzschild radius R.

Schwarzschild Trumpet Data

We make a standard 3+1 decomposition of the black hole metric,

$$ds^2 = -\alpha^2 dt^2 + \gamma_{ij}(dx^i + \beta^i dt)(dx^j + \beta^j dt), \quad (1)$$

where α is the lapse function which describes how time flows between the space-like slices, γ_{ij} is the 3-metric on a slice and β^i is the shift vector which describes how the points on each slice move from one slice to the next.



In the moving-puncture method, wormhole black hole initial data relax to trumpet coordinates, as in Figure 1. Ideally, we would use initial data in trumpet coordinates, but we only have trumpet data for a stationary Schwarzschild black hole [1]. The goal here is to construct trumpet initial data for a boosted black hole.

Constructing Advecting Axisymmetric Trumpet Data

The geometry of how the slices are embedded in the 4-D manifold is described by the extrinsic curvature, K_{ij}, which along with the metric, makes up the data which fully describes each slice $\{\gamma_{ij}, K_{ij}\}$. For convenience this can be decomposed into its trace K and trace-free A_{ij} parts. In the conformal thin sandwich (CTS) formalism, we define a conformal factor ψ such that A_{ij} can be written in terms of the conformal extrinsic curvature $\tilde{A}_{ij} = \psi^2 A_{ij}$.

In terms of $\{\tilde{\gamma}_{ij}, \partial_t \tilde{\gamma}_{ij}, K, \partial_t K\}$, $\tilde{A}_{ij}$ can be written,

$$\tilde{A}^{ij} = \frac{1}{2\tilde{\alpha}}((L\beta)^{ij} - \partial_t \tilde{\gamma}^{ij}). \quad (2)$$

Here we choose maximal slicing condition $K=0$, and impose an advective condition $\partial_t f = -v\partial_z f$ so that the puncture is moving with constant velocity v in the z direction. As a first guess we start with a flat metric $\tilde{\gamma}^{ij} = \delta^{ij}$ and $\tilde{A}^{ij} = \tilde{A}^{ij}_{BY} + \tilde{A}^{ij}_{T}$ where BY denotes the Bowen-York extrinsic curvature and T denotes the Trumpet coordinate extrinsic curvature. We must now solve the constraint equations iteratively for the remaining data, ψ, β^i, $\tilde{\alpha}$.

$$\tilde{A}^{ij} = \tilde{A}^{ij}_{BY} + \tilde{A}^{ij}_{T}$$

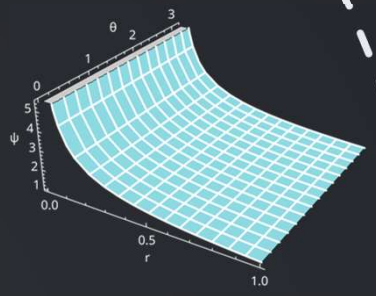
The Hamiltonian Constraint

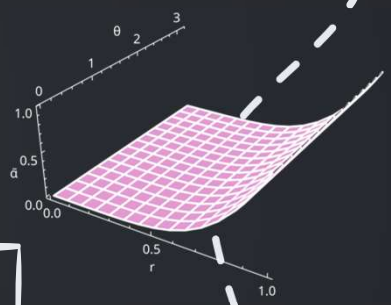
Throughout, we work in spherical polar coordinates, (t, r, θ, φ) with axisymmetry in φ so that our spatial slices are 2-dimensional in r and θ, and the puncture maps to the boundary along $r = 0$ for $\theta \in [0, \pi)$.

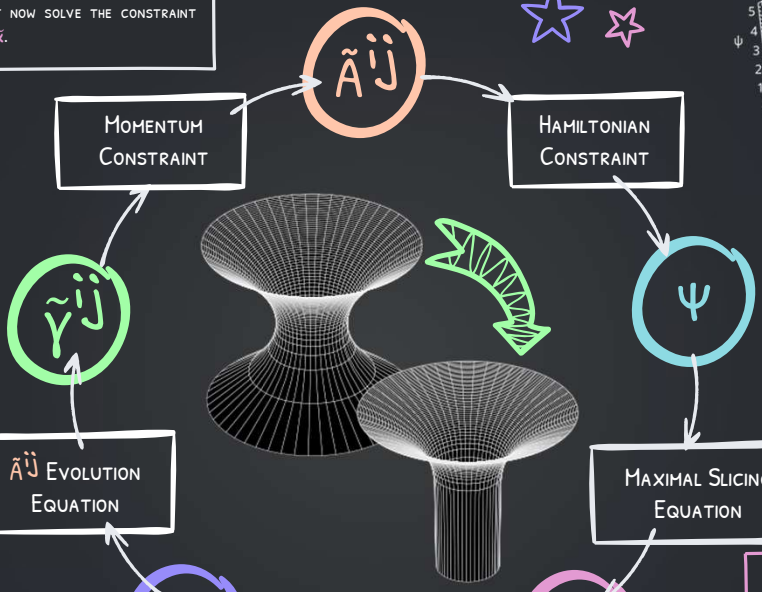
The first step is to solve the Hamiltonian constraint for the Bowen-York correction to the conformal factor. For the maximally sliced case, the Hamiltonian constraint is,

$$\tilde{\nabla}^2 \psi + \frac{1}{8}\psi^{-7}\tilde{A}^{ij}\tilde{A}_{ij} = 0. \quad (3)$$

This is a nonlinear equations which can be linearised and solved iteratively. The solution is given below.







Extrinsic Curvature Evolution Equation

In order to solve for $\tilde{\gamma}^{ij}$ we make use of the metric reference form of the Einstein field equations used by Cook and Baumgarte 2008 [3]. This provides us with an elliptic equation in $\tilde{\gamma}^{ij}$ with all of the source terms calculated in terms of the flat metric.

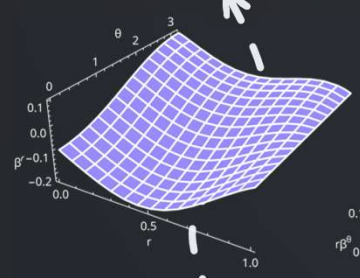
Once we have updated the metric so it is no longer conformally flat, we can solve the momentum constraint to calculate a correction to the original Bowen-York and trumpet extrinsic curvatures. We repeat the process, and iterate until we get a solution within our error tolerance.

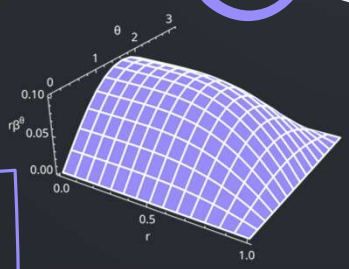
Maximal Slicing Condition

Now we have the Bowen-York Trumpet conformal factor we can compute $\tilde{\alpha}$ by solving the K evolution equation which can be simplified to the maximal slicing equation:

$$\tilde{\nabla}^2(\psi^7\tilde{\alpha}) - \frac{7}{8}\psi^{-8}\tilde{A}^{ij}\tilde{A}_{ij}(\psi^7\tilde{\alpha}) = 0, \quad (4)$$

which is linear in $\psi^7\tilde{\alpha}$, and for our boosted trumpet application, $\tilde{\alpha}$ is regular at the puncture. The solution is above in pink.



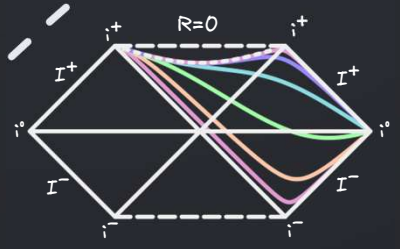


Solving for β^i

Given the extrinsic curvature, we can calculate the shift following [2] by taking the divergence of Equation (2),

$$\tilde{\nabla}_j(2\tilde{\alpha}\tilde{A}^{ij}) = \tilde{\nabla}_j(L\beta)^{ij}. \quad (5)$$

The solutions are above in purple.



References

[1] Hannam, M., Husa, S., Ohme, F., Brügmann, B., & O'Murchadha, N. 2008, Wormholes and trumpets: the Schwarzschild spacetime for the moving-puncture generation.
[2] Tichy, W., Brügmann, B., & Laguna, P. 2003, Physical Review D, 68.
[3] Cook, G. B., & Baumgarte, T. W. 2008, Physical Review D, 78.